FDS EXPERIMENT-11

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Experiment to understand pandas library using use cases in data science.

```
import numpy as np
import pandas as pd
```

```
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya", "Ezhil"],
    "Age": [20, 21, 19, 22, 20],
    "Marks": [85, 90, 75, 95, 80]
}

df = pd.DataFrame(data)

print(df)
```

```
    Name  Age  Marks
0   Arun   20     85
1   Bala   21     90
2  Charan   19     75
3  Divya   22     95
4  Ezhil   20     80
```

```
print(df.head(3))
print(df.tail(2))
print(df.describe())
```

```
    Name  Age  Marks
0   Arun   20     85
1   Bala   21     90
2  Charan   19     75
    Name  Age  Marks
3  Divya   22     95
4  Ezhil   20     80
             Age      Marks
count   5.000000   5.000000
mean   20.400000  85.000000
std     1.140175   7.905694
min    19.000000  75.000000
25%    20.000000  80.000000
50%    20.000000  85.000000
75%    21.000000  90.000000
max    22.000000  95.000000
```

```
print("Student Names:")
print(df["Name"])

print("Students scoring above 80:")
print(df[df["Marks"] > 80])
```

```
Student Names:
0      Arun
1      Bala
2    Charan
3     Divya
4     Ezhil
Name: Name, dtype: object
Students scoring above 80:
    Name  Age  Marks
0   Arun   20     85
1   Bala   21     90
3  Divya   22     95
```

```python
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya"],
    "Marks": [85, None, 75, 95]
}

df = pd.DataFrame(data)

print("Missing Values:")
print(df.isnull().sum())

df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

print("After Filling Missing Values:")
print(df)
```

```
Missing Values:
Name     0
Marks    1
dtype: int64
After Filling Missing Values:
     Name  Marks
0    Arun   85.0
1    Bala   85.0
2  Charan   75.0
3   Divya   95.0
```

```
print("Total Marks:", df["Marks"].sum())
print("Average Marks:", df["Marks"].mean())
print("Maximum Marks:", df["Marks"].max())
print("Minimum Marks:", df["Marks"].min())
print("Number of Students:", df["Name"].count())
```

```
Total Marks: 340.0
Average Marks: 85.0
Maximum Marks: 95.0
Minimum Marks: 75.0
Number of Students: 4
```

```
data = {
    "Department": ["CSE", "ECE", "CSE", "ECE", "CSE"],
    "Marks": [85, 75, 90, 80, 95]
}

df = pd.DataFrame(data)

result = df.groupby("Department")["Marks"].mean()

print(result)
```

```
Department
CSE    90.0
ECE    77.5
Name: Marks, dtype: float64
```